# Synergy Identification via Network Clustering

## - A Model-Agnostic Framework for Discovering Global Feature Interactions


Simon Bock Segaard
USC Ming Hsieh Department of Electrical and Computer Engineering
Aalborg University Department of Mathematical Sciences


AALBORG UNIVERSITY

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## Introduction

**The Black-Box Challenge**

Modern machine learning models achieve exceptional performance across diverse fields, yet they remain "black-boxes". While we can measure individual feature importance using tools like SHAP [1] and find pairwise interaction with Friedman's H-statistic [2], discovering feature interactions used by a model is still a significant challenge. Existing methods are limited to local insight and fail to capture the global architectural logic of the model [3].

## Methodology

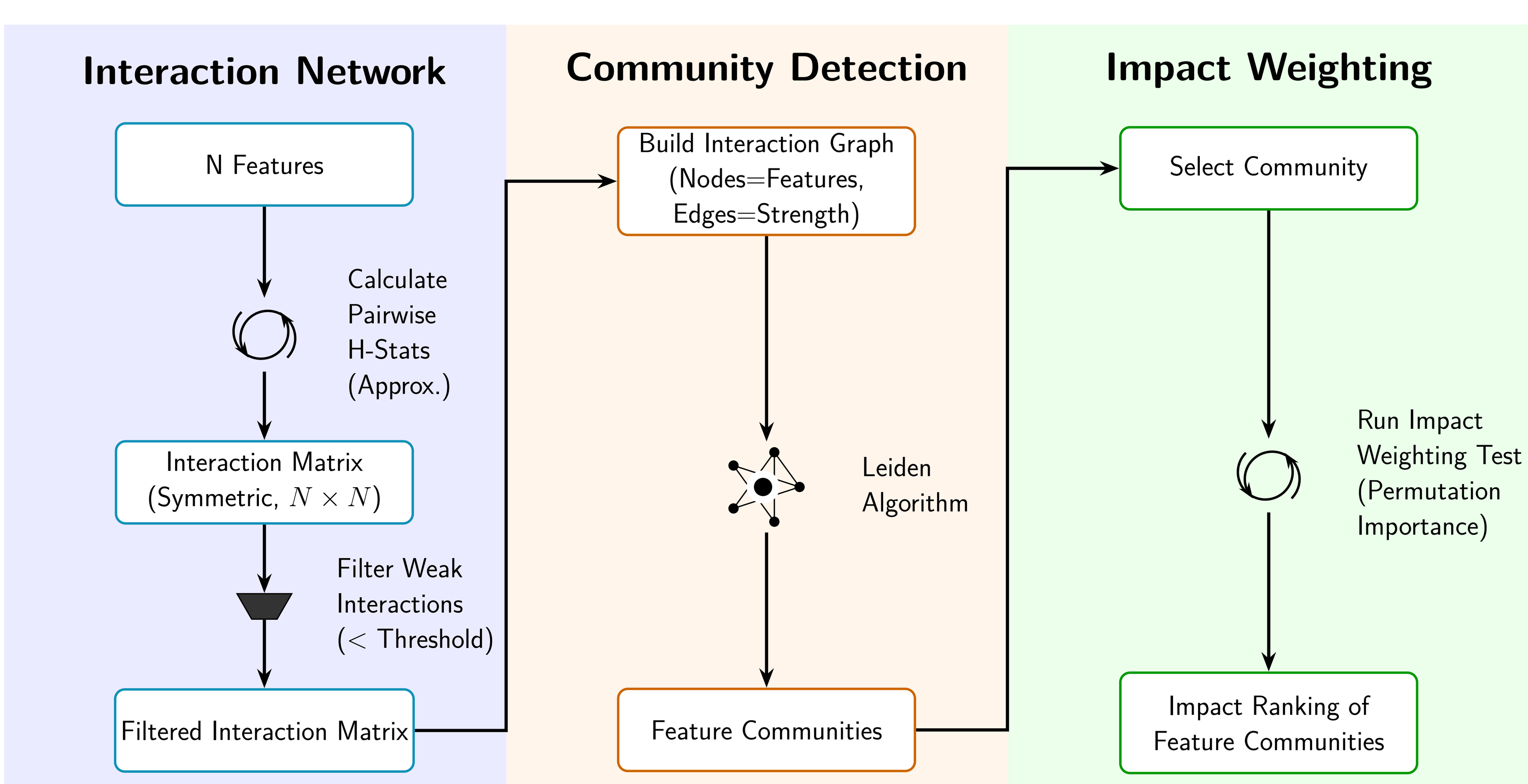
**The SINC Pipeline**

We introduce **Synergy Identification via Network Clustering (SINC)**, a fully model-agnostic framework designed to decode how a model views its input features by finding global interactions. SINC consists of three phases:

I. Interaction Network:
   - An interaction network is created where edges represent pairwise H-statistic scores that exceed a defined cutoff $\tau$.

II. Community Detection:
   - Using the Leiden Algorithm [4] iteratively with the Constant Potts Model as the quality function the features are sorted into communities.

III. Impact Weighting:
   - The impact of each community on the model's performance is quantified using permutation-based importance.




## Application & Future Work

**Application**

- When paired with a high-performing model, SINC can illuminate interactions between multiple features in complex data.
- SINC can be used to debug underperforming models if there is known interactions in the data.

**Future Work**

- Further validate SINC, both empirically and analytically.
- Apply SINC to models trained on complex datasets, such as StressID.


Contact: ah_235@usc.edu, jm_496@usc.edu, ss_768@usc.edu

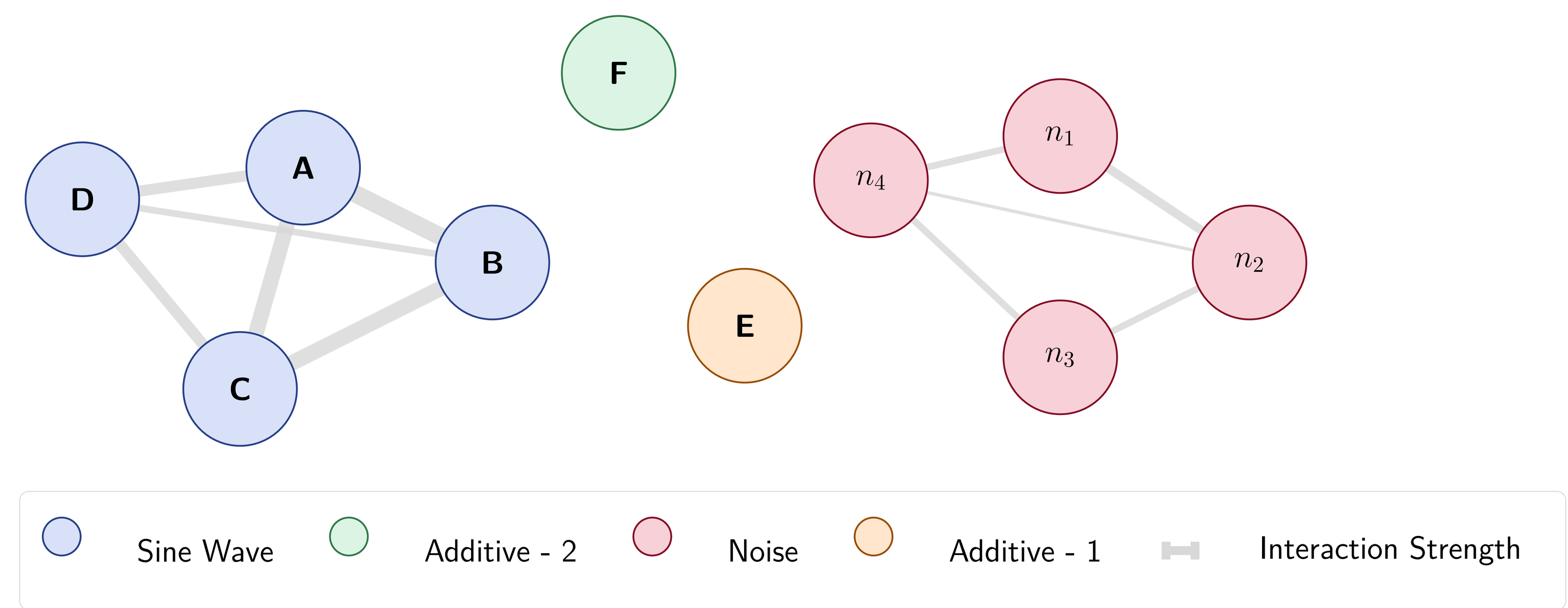

## Experimental Analysis

**Simulated Data**

- The data is created using 6 features (A, B, C, D, E, F) via the following equation:
  $y = 20 \cdot sin(\pi \cdot A \cdot B \cdot C \cdot D) + 10 \cdot E - 5 \cdot F + \mathcal{N}(0, 0.1)$
- In addition to the 6 features each datapoint includes 4 redundant noise features.
- SINC is used on a Random Forrest Regressor with high accuracy trained on the simulated data.

**Feature Interaction Network Created by SINC**

Simulated Data




| Community | Size | Impact |
|---|---|---|
| Sine Wave | 4 | 32.447537 |
| Additive – 1 | 1 | 13.177714 |
| Additive – 2 | 1 | 2.761918 |
| Redundant Features | 4 | 0.028036 |

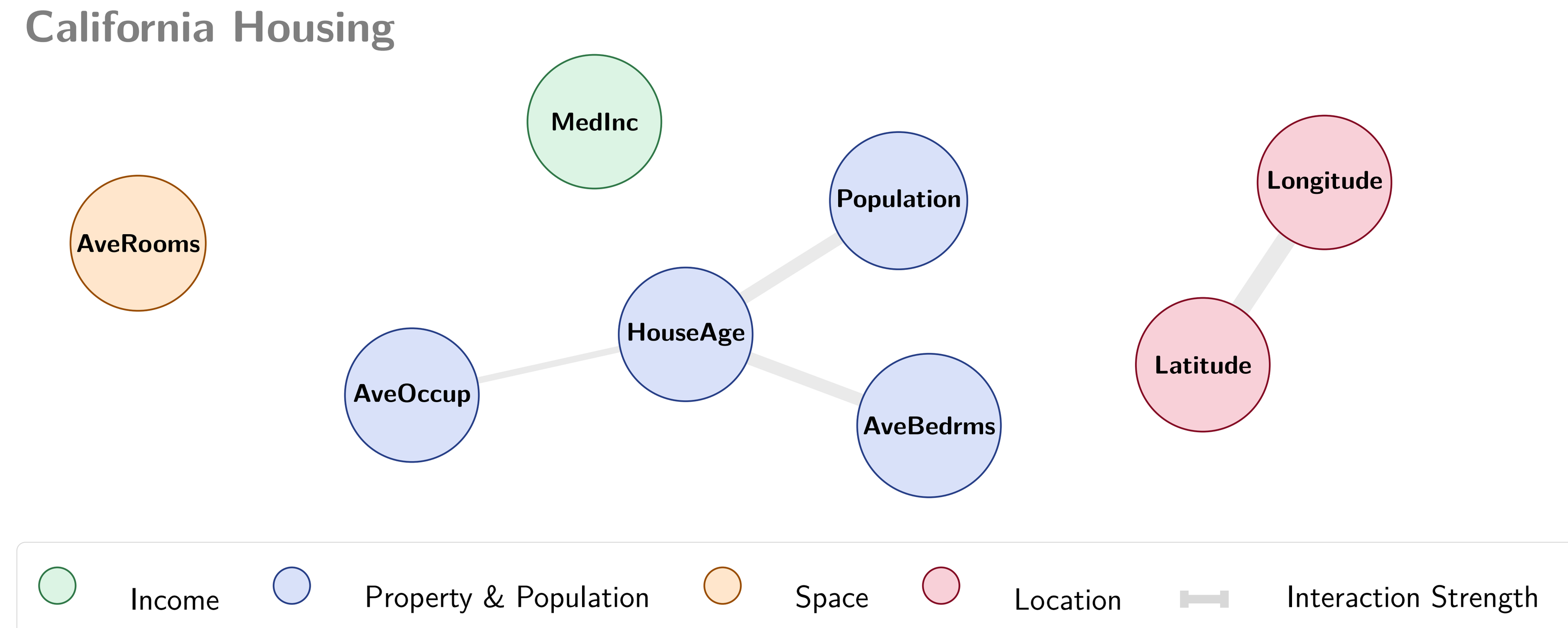
- SINC shows that the model understands how to group the features and that the groups impact match what we would expect.

**California Housing**

- Contains 8 features and the goal is to predict the price of homes in California.
- The longitude, latitude, and median income is the most important, however longitude and latitude only makes sense together.
- SINC is used on a high performing Histogram-based Gradient Boosting Regression Tree.

**Feature Interaction Network Created by SINC**

California Housing




| Community | Size | Impact |
|---|---|---|
| Location | 2 | 0.692148 |
| Income | 1 | 0.581503 |
| Property & Population | 4 | 0.247835 |
| Space | 1 | 0.090343 |

- As excepted, since the model performs well, it finds and utilizes the interaction between latitude and longitude correctly.
- The impact scores also match what we expect.


**References**

[1]Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In Advances in Neural Information Processing Systems (Vol. 30). Curran Associates, Inc.

[2]Friedman, J. H., & Popescu, B. E. (2008). Predictive learning via rule ensembles. The Annals of Applied Statistics, 2(3), 916–954. https://doi.org/10.1214/07-AOAS148

[3]Tsang, M., Rambhatla, S., & Liu, Y. (2020). How does this interaction affect me? Interpretable attribution for feature interactions. In Advances in Neural Information Processing Systems (Vol. 33). Curran Associates, Inc.

[4]Traag, V. A., Waltman, L., & van Eck, N. J. (2019). From Louvain to Leiden: Guaranteeing well-connected communities. Scientific Reports, 9, 5233. https://doi.org/10.1038/s41598-019-41695-z